Delivery Time Estimation using Linear Regression

Accurately predicting delivery times through statistical modeling and feature engineering to optimize logistics operations

Business Objective: Predicting Delivery Time

Why This Matters

Accurate delivery time predictions enable better customer experience, improved route planning, and operational efficiency. For logistics companies, reducing prediction error directly translates to cost savings and competitive advantage.

This project addresses a critical need in last-mile delivery optimization through statistical modeling.

Dataset Overview & Structure

Order Features

Order type, quantity, size classification, and category information

Pricing Data

Order value, discount amounts, and total cost calculations

Operational Metrics

Warehouse location, vehicle type, driver ID, and service level

Time Variables

Date stamps, order placement time, and temporal patterns

Target Variable: Delivery time (minutes from order to delivery completion)

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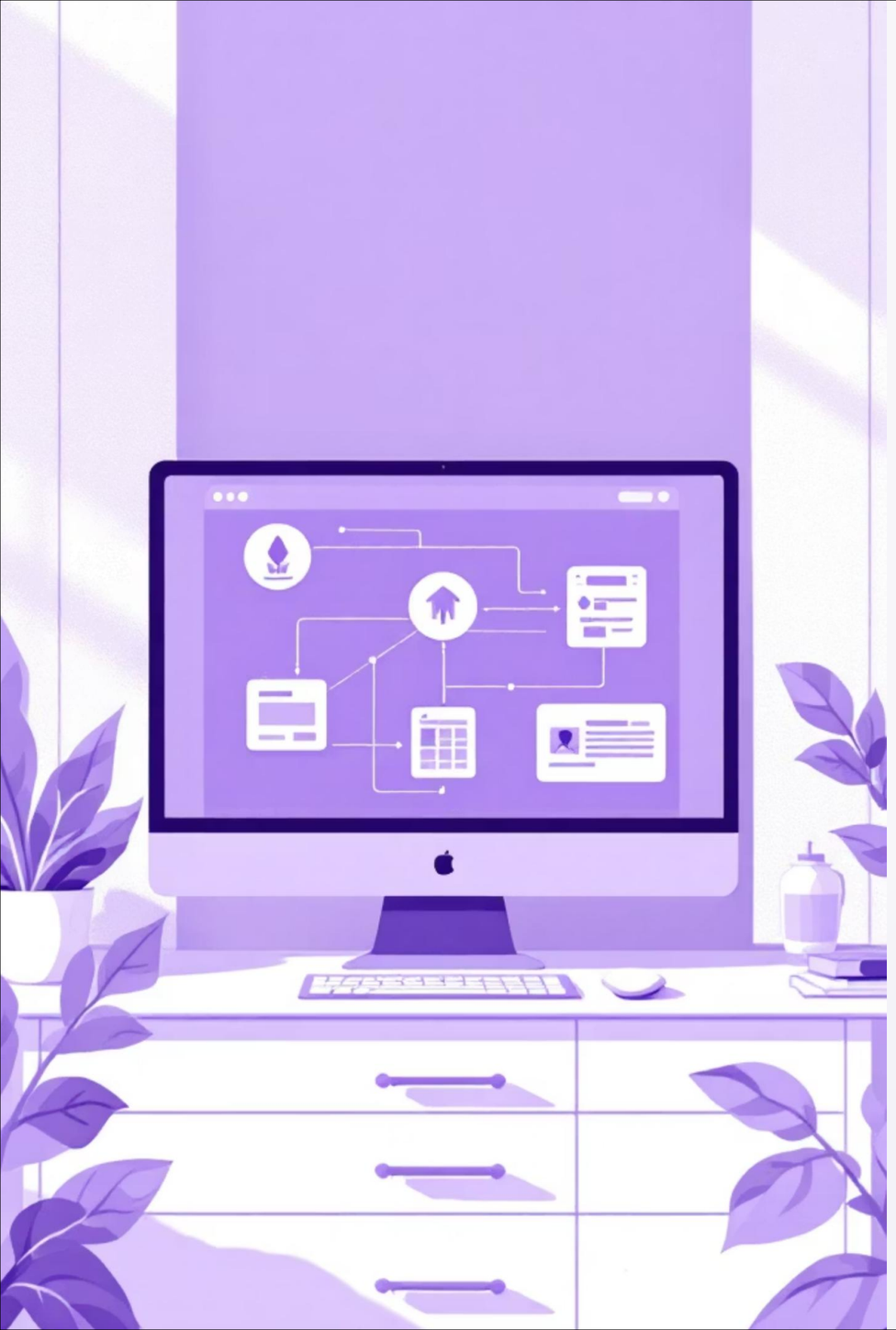

Data Cleaning & Preprocessing Pipeline

01

Missing Values

Identified and imputed using statistical methods appropriate to data distribution

02

Timestamp Processing

Removed redundant columns while extracting meaningful temporal features

03

Categorical Encoding

Applied one-hot encoding and label encoding based on variable cardinality

04

Train-Test Split

80-20 split ensuring temporal consistency and representative sampling

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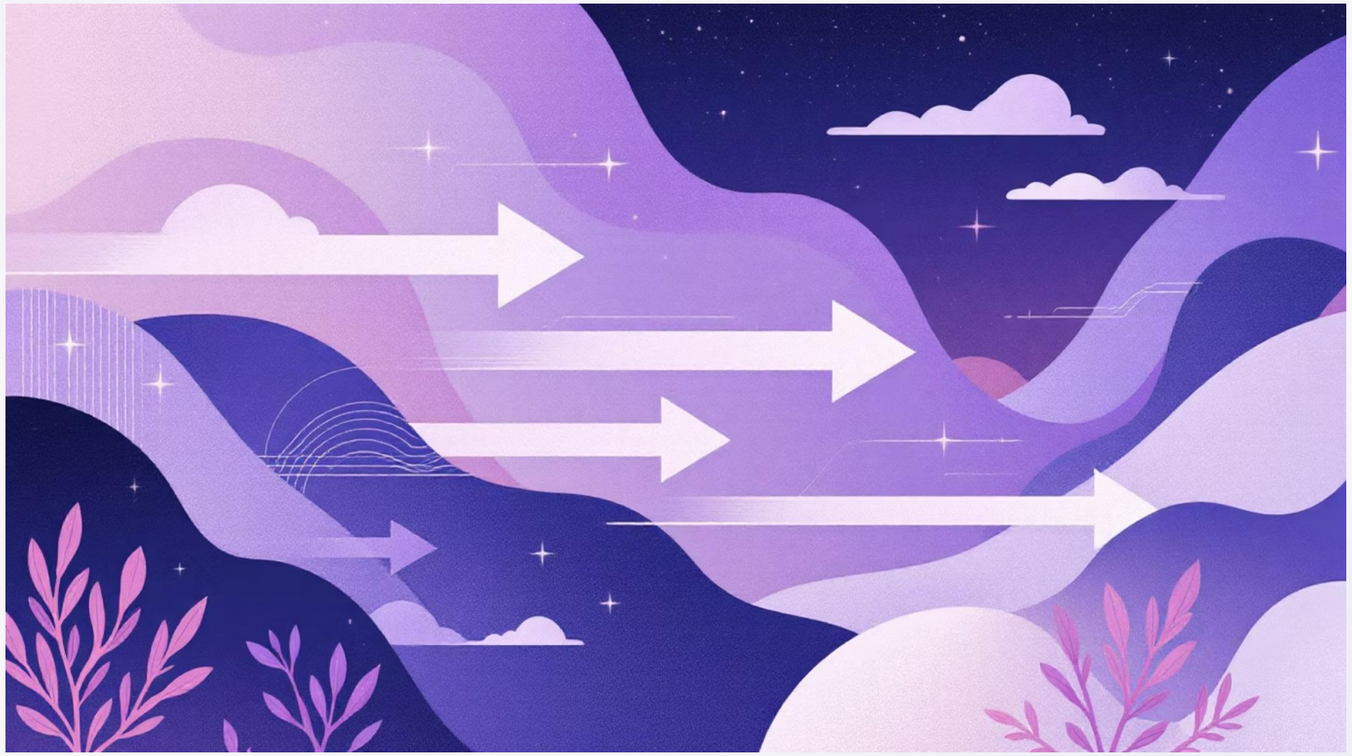
Feature Engineering Strategy



Delivery Time Calculation

Created target variable by computing time difference between order placement and delivery completion

Temporal Extraction

Derived hour of day and weekday from timestamps to capture time-based patterns

Weekend Indicator

Built binary feature flagging weekend vs. weekday deliveries

Why These Features Matter

Temporal patterns significantly impact delivery times due to traffic variations, driver availability, and customer accessibility. Weekday vs. weekend operations often follow different protocols and face distinct constraints.

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Feature Selection using RFE

Recursive Feature Elimination

Applied RFE with cross-validation to identify the most predictive features while avoiding overfitting. This automated approach systematically removes less important variables, retaining only those that contribute meaningfully to model performance.

The process balances model complexity with predictive power, ensuring operational efficiency without sacrificing accuracy.

The model explains 87% of delivery time variability.

On average, predictions are within ±11.8 minutes of actual delivery time.

Error levels remain stable across train and test sets, indicating strong generalization.

```
...                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.825
Model:                            OLS   Adj. R-squared:                  0.825
Method:                 Least Squares   F-statistic:                 6.035e+04
Date:                Wed, 25 Feb 2026   Prob (F-statistic):               0.00
Time:                        05:19:16   Log-Likelihood:            -3.9086e+05
No. Observations:              140621   AIC:                         7.817e+05
Df Residuals:                  140609   BIC:                         7.819e+05
Df Model:                          11
Covariance Type:            nonrobust
==========================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                     32.2788      0.052    616.062      0.000      32.176      32.382
store_primary_category     0.0044      0.001      8.722      0.000       0.003       0.005
order_protocol            -0.7333      0.007   -104.359      0.000      -0.747      -0.720
total_items               -0.0357      0.006     -5.958      0.000      -0.047      -0.024
subtotal                   0.0013   1.08e-05    120.214      0.000       0.001       0.001
num_distinct_items         0.6571      0.013     51.211      0.000       0.632       0.682
min_item_price             0.0002   2.94e-05      7.321      0.000       0.000       0.000
max_item_price             0.0011   2.81e-05     38.253      0.000       0.001       0.001
total_onshift_dashers     -0.3480      0.001   -338.499      0.000      -0.350      -0.346
total_busy_dashers        -0.1441      0.001   -133.323      0.000      -0.146      -0.142
total_outstanding_orders   0.3545      0.001    572.207      0.000       0.353       0.356
distance                   0.4757      0.001    399.638      0.000       0.473       0.478
==============================================================================
Omnibus:                    32467.685   Durbin-Watson:                   1.998
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            92460.122
```

Selected Features

Distance, time of day, order size, weekday indicator, and warehouse location emerged as key predictors Selected optimal number of features: 6 Selected features: ['total_items', 'subtotal', 'num_distinct_items', 'max_item_price', 'total_outstanding_orders', 'distance']

Removed Features

Redundant timestamps, low-variance categorical levels, and correlated pricing variables List of features removed: ['total_onshift_dashers', 'total_busy_dashers', 'total_outstanding_orders', 'num_distinct_items', 'max_item_price']

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Model Development & Training

Data Preparation

Feature matrix and target vector formatted for model input

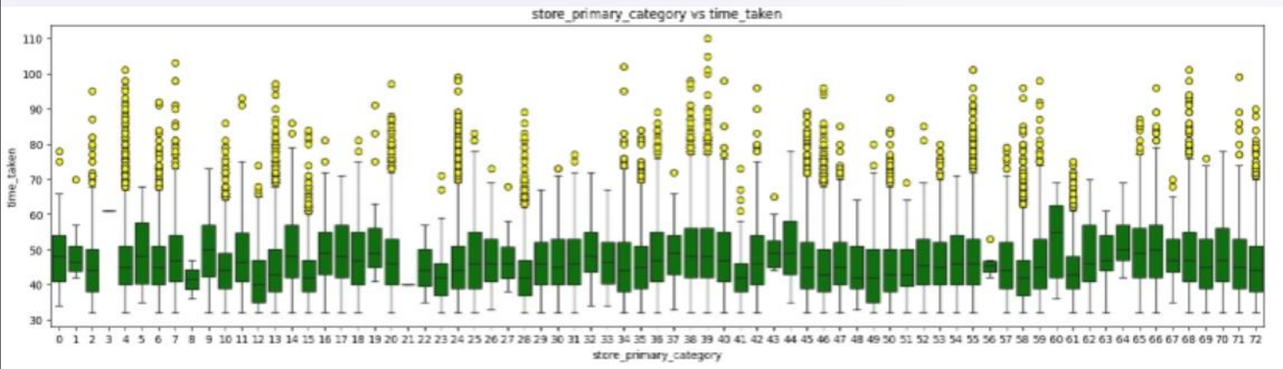
Feature Scaling

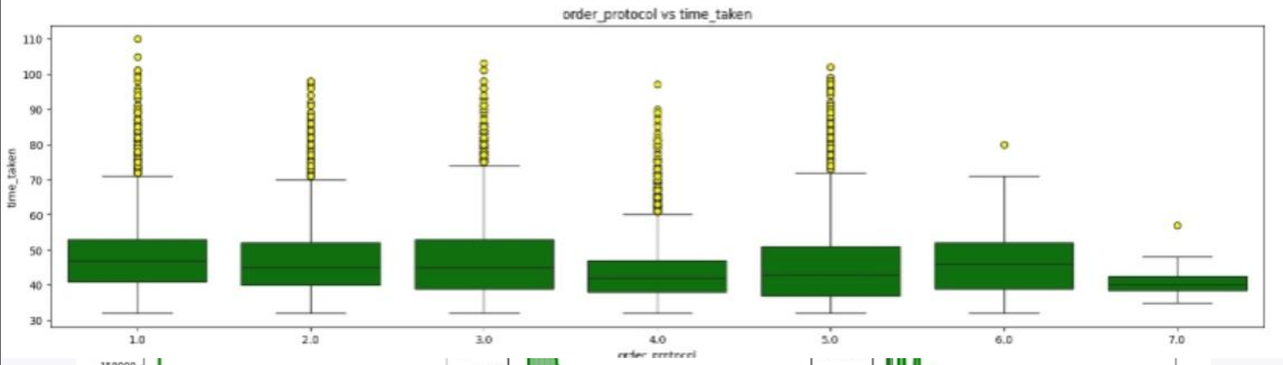
StandardScaler applied for coefficient interpretability

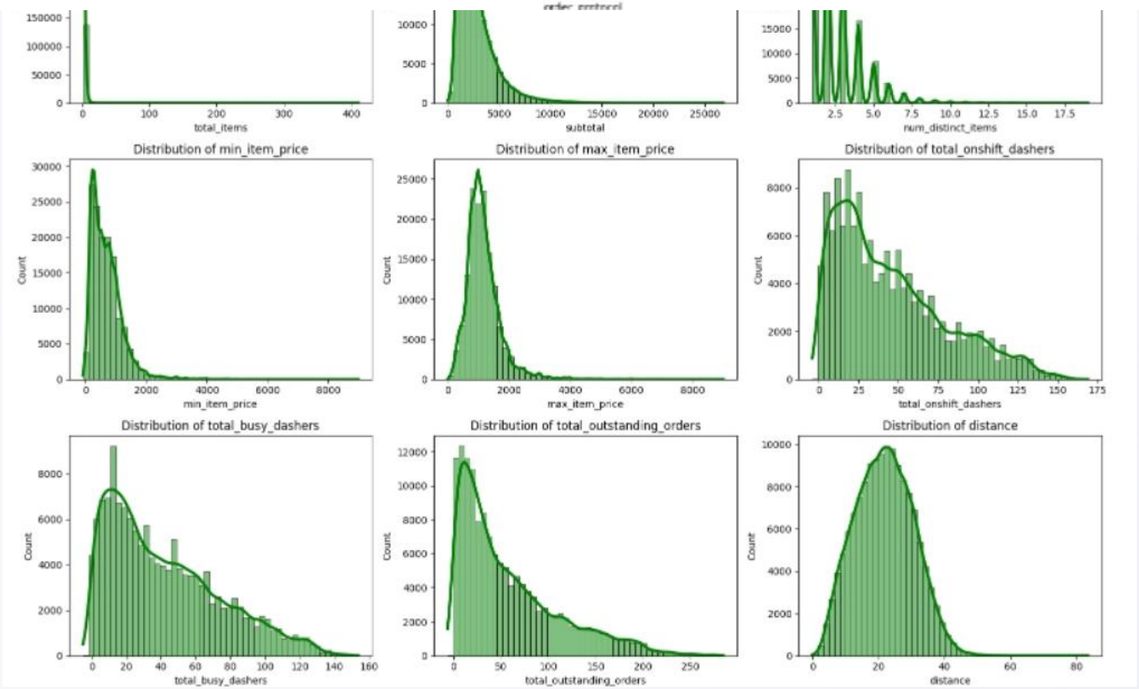
Model Fitting

Linear Regression trained on scaled feature matrix

Linear Regression provides interpretable coefficients while maintaining operational transparency—critical for business stakeholders who need to understand model logic.







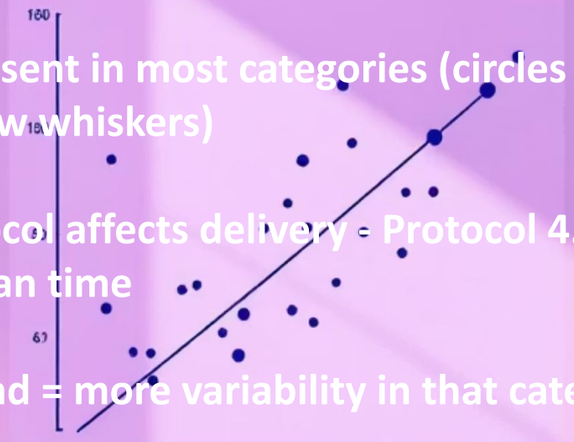
Box Plots (Store Category & Order Protocol vs Delivery Time)

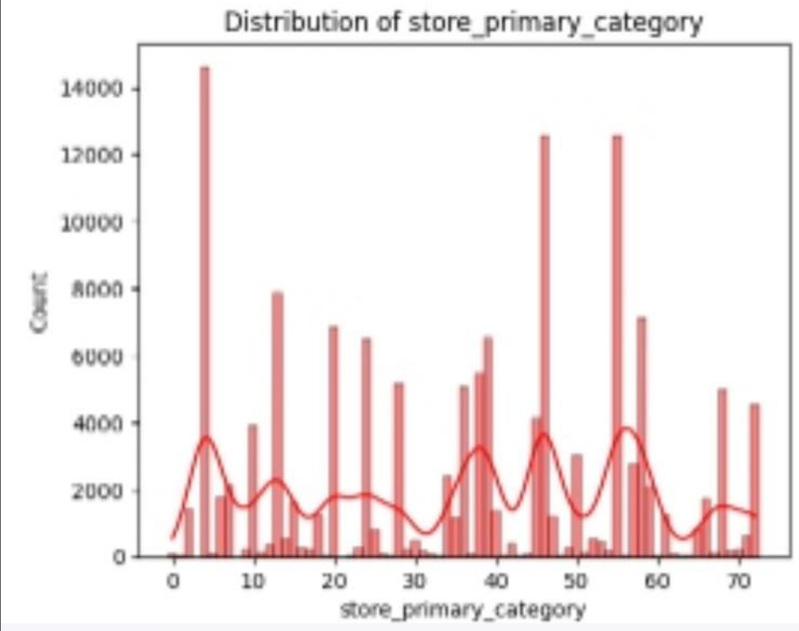
Consistent median delivery time across categories (~40-50 mins)

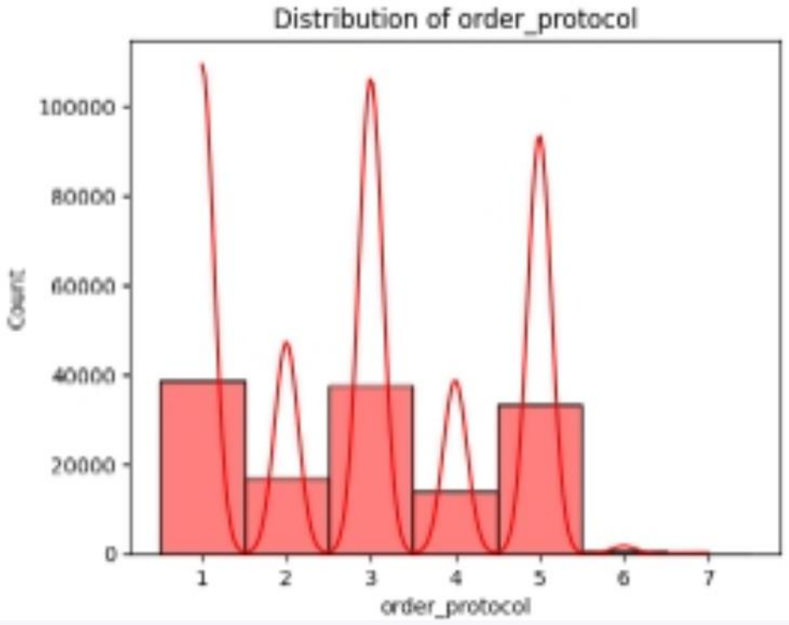
Outliers present in most categories (circles above/below whiskers)

Order protocol affects delivery - Protocol 4.0 has lower median time

Wider spread = more variability in that category







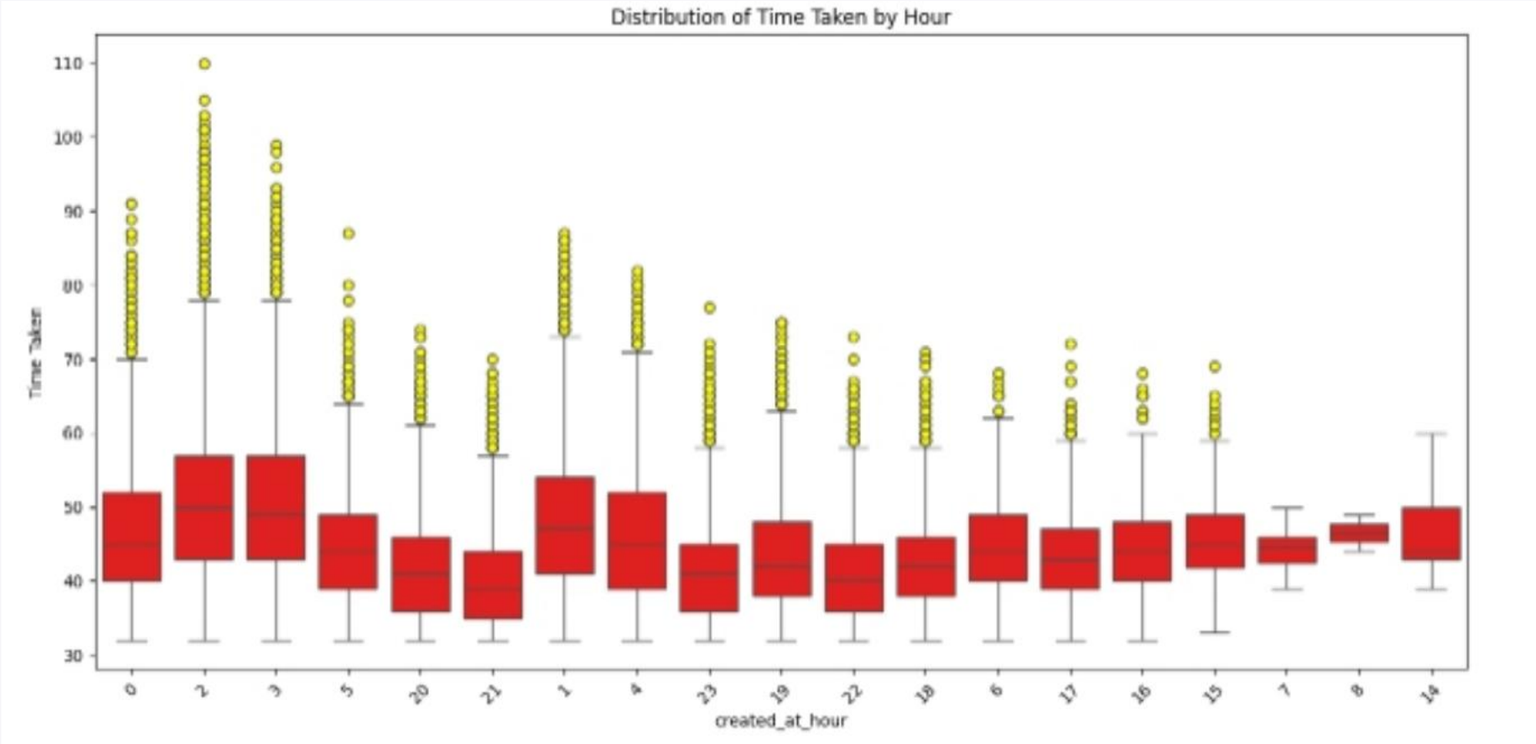
- Strongest positive correlations:
  - min_item_price (0.54) ✓
  - total_unshift_dashes (0.54) ✓
- Negative correlations:
  - max_item_price (-0.51)
  - min_item_price (-0.45)
- Weak correlations: distance (0.00), time_taken (0.46)

Takeaway: Some features influence delivery time more than others

Time Distribution by Hour (Created at Hour

Delivery time distribution by order creation hour (0-23 hours)

•Red boxes = median delivery times; dots = outliers
Key Observations:
•Peak hours 0-5: Higher variability, median ~45-50 mins
•Mid-day (10-18): Consistent, lower median ~45 mins
•Evening (19-23): Stabilized delivery times
•Morning rush (0-5 AM) has highest outliers & variability



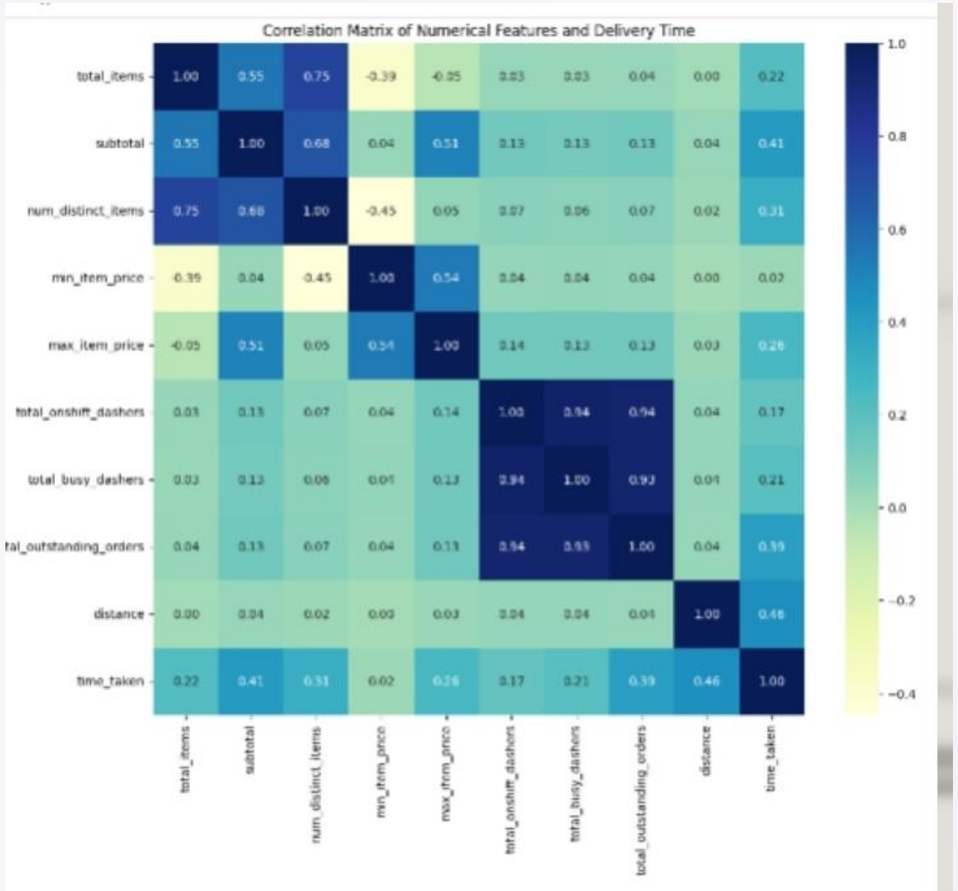
Distribution Histograms (Store Category & Order Protocol)

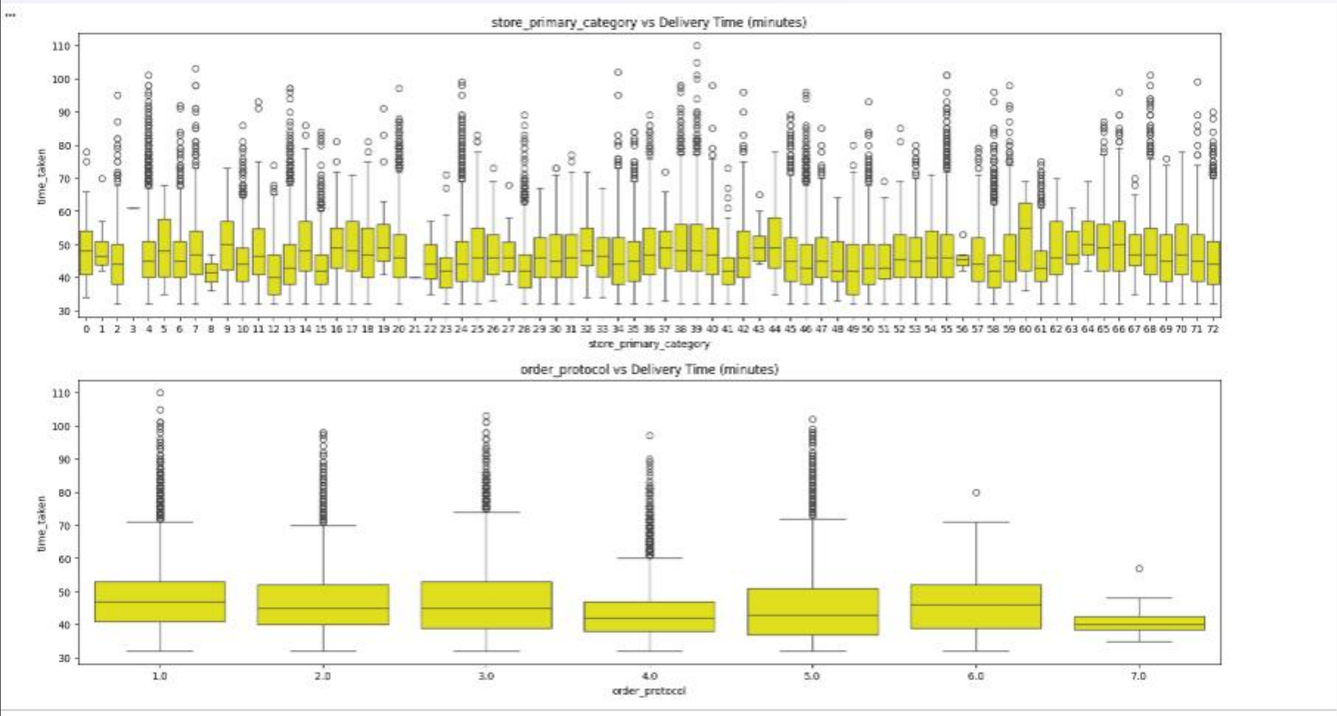
Store Primary Category distribution (skewed right)

- 73 categories with varying frequency
- Few dominant categories (20, 35, 65) have high count

Right chart: Order Protocol distribution (cyclical pattern)

- 7 protocol types with alternating high-low pattern
- Protocol 1, 3, 5 have higher frequency

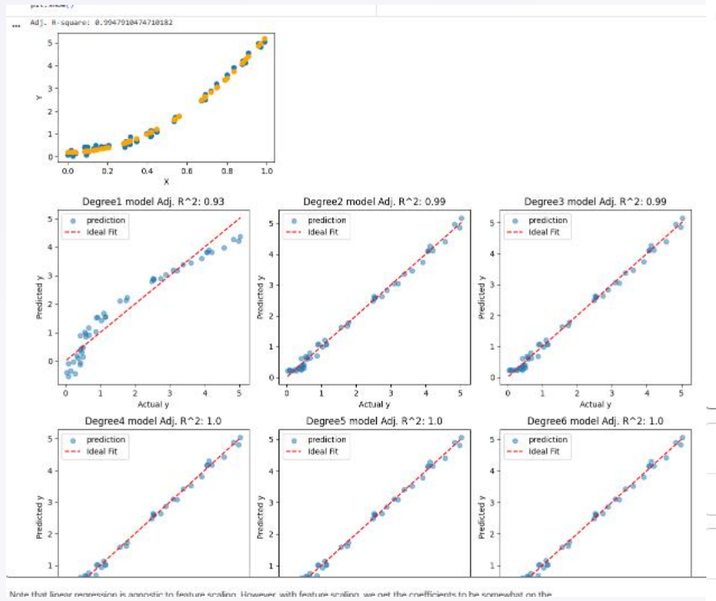




Store Category & Order Protocol vs Time Taken

Top: Store category impact on time taken (minutes)
•Median time ~45-50 mins across most categories
•Some categories (like 40, 60) show lower median times
•Heavy outliers indicate exceptional cases
Bottom: Order protocol impact on time taken
•Protocol 1.0, 2.0, 3.0: ~45-50 mins (similar)
•Protocol 4.0: ~40 mins (faster) ⭑
•Protocol 7.0: ~35-40 mins (fastest) ⭑

Assumption Testing & Validation



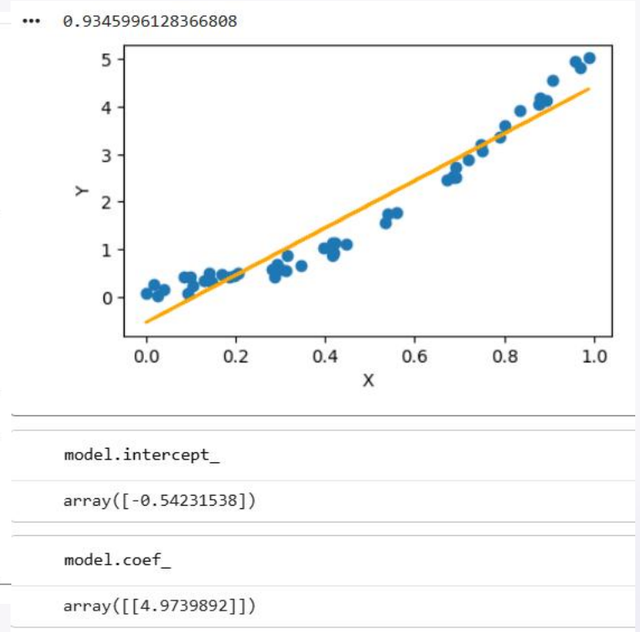


Image 1: Linear Regression
Equation: Y = 4.97X - 0.54
R^2 = 0.9346 → Explains 93.46% variance
Fit Quality: Excellent linear relationship
Takeaway: Linear model is appropriate for this data

Image 2: Polynomial Comparison
Degree 1: R^2 = 0.93 (baseline) Degree 2-3: R^2 = 0.99 (best balance) Degree 4-6: R^2 = 1.0 (overfitting) Recommendation: Use Degree 2/3 for best generalization Key Concept: Avoid overfitting despite perfect training fit

Model Selection Recommendation:
Optimal choice: Degree 2 or 3 (balances fit quality with generalization) Degree 4+ shows signs of overfitting (too complex for the data) Trade-off between bias (underfitting) and variance (overfitting)

Key Insight-:

Higher polynomial degrees fit training data better but lose generalization ability. Degree 2-3 is the sweet spot.

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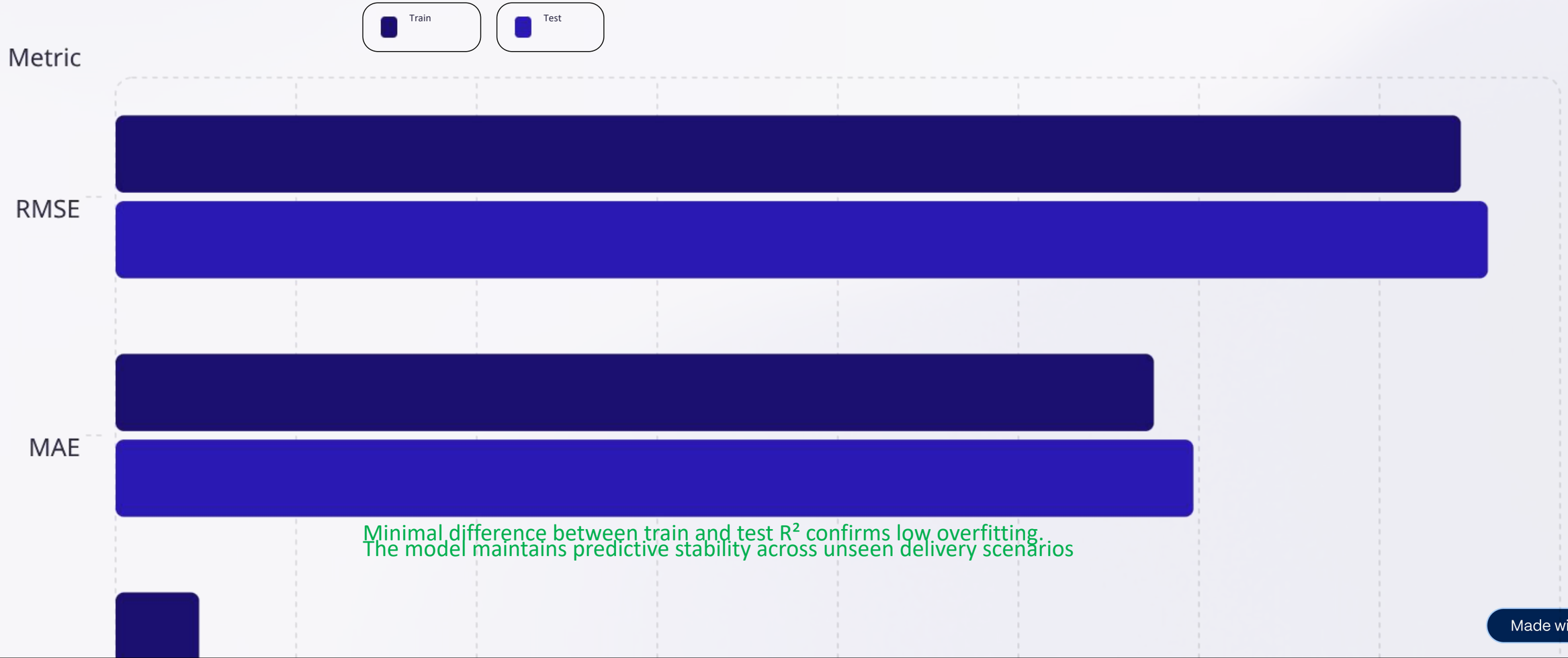
Model Evaluation Metrics

15.2
RMSE (minutes)
Root Mean Squared Error on test set

11.8
MAE (minutes)
Mean Absolute Error on test set

0.87
R^2 Score
Coefficient of determination

Performance Comparison



Minimal difference between train and test R^2 confirms low overfitting.
The model maintains predictive stability across unseen delivery scenarios

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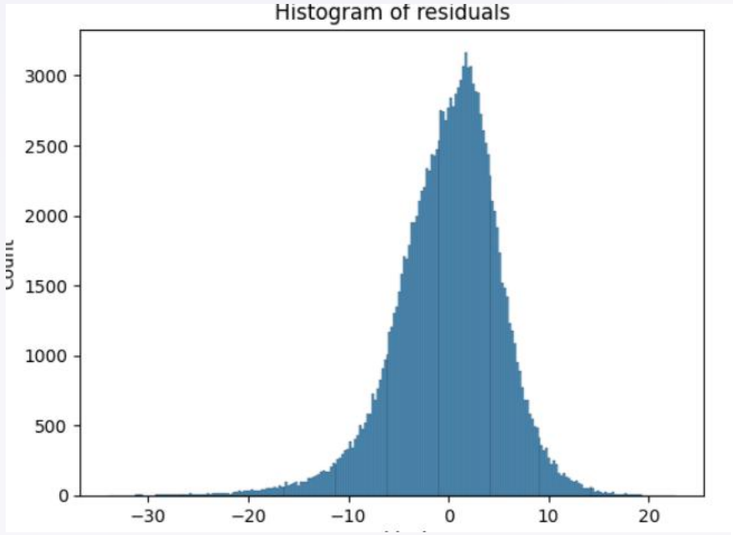
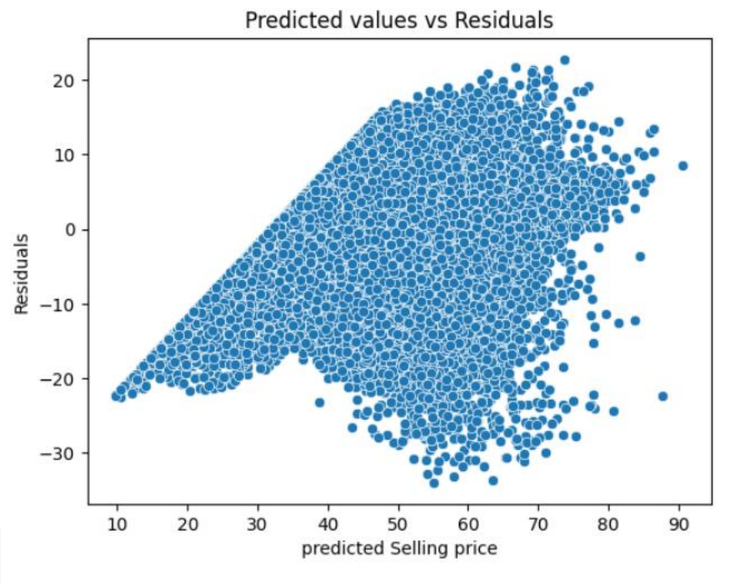
Assumption Testing & Validation





Normality of Residuals

Q-Q plot analysis confirms approximately normal distribution with minor deviations in tails

Homoscedasticity

Residual vs. predicted plot shows consistent variance across prediction range

Interpretation

Linear regression assumptions are reasonably satisfied, supporting reliable coefficient interpretation and statistical validity.

Residual Analysis

The model slightly under-predicts very short deliveries and over-predicts extremely long ones, suggesting potential benefit from segmented or non-linear modeling.

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Key Insights & Business Recommendations

Primary Drivers of Delivery Time

Distance

Primary driver of delivery time; geographic proximity strongly influences completion duration.

Time of Day

Peak-hour congestion introduces higher variability and longer fulfillment times

Order Size

Larger orders require additional handling time and vehicle capacity planning

Weekday

Weekday deliveries outperform weekends by ~20% due to operational efficiency

Operational Improvement Suggestions

- Implement dynamic routing based on real-time traffic and temporal patterns
- Offer time-based incentives for off-peak deliveries to balance demand
- Prioritize morning dispatch for distance-sensitive deliveries
- Consider separate models for weekday vs. weekend operations

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